AI Smart Well

Platform Functional Overview

AI-powered SaaS platform for oil & gas well analysis, optimization, and Slot Perforation Technology (SPT) evaluation

30+	9	14	12
Functional Modules	Analysis Stages	Database Tables	Backend Functions

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1. Executive Summary

AI Smart Well is a multi-tenant SaaS platform designed to revolutionize well evaluation and optimization in the oil and gas industry. The platform combines advanced AI analytics, computer vision, and proprietary Slot Perforation Technology (SPT, US Patent 8,863,823) to identify bypassed reserves and maximize production from existing wellbores.

Figure 1 — Dashboard: Portfolio KPIs and Platform Modules

Core Value Proposition

- Automated 9-stage well analysis pipeline — from field scanning to EOR optimization
- AI-powered geological interpretation using NVIDIA NIM computer vision
- Quantum-enhanced Monte Carlo simulations for economic risk analysis
- Real-time data integration from state registries (TX, OK, KS, NM, CO, ND, WY)
- Multi-tenant architecture with Row-Level Security (RLS) data isolation
- Patented SPT technology as primary EOR recommendation engine

Target Market

- Independent operators managing 50–5,000+ wells
- Companies seeking to revitalize low-production (<10 bbl/d) well portfolios
- Engineering firms conducting reservoir evaluation studies

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2. Core 9-Stage Analysis Pipeline

The platform's backbone is a standardized 9-stage well analysis pipeline that takes a candidate well from initial screening through full economic optimization.

Stage 1 Field Scanning
Regional scanning to identify low-production wells (<10 bbl/d). Filters by state, county, operator, formation.

Stage 2 Data Classification
Automated data quality scoring and completeness assessment. AI auto-fills missing parameters.

Stage 3 Core Analysis
AI-powered rock sample interpretation via NVIDIA NIM computer vision. Three CV modes.

Stage 4 Cumulative Analysis
Production decline curve analysis using Arps methodology (b=0.5). IOIP and recovery factor.

Stage 5 Seismic Reinterpretation
CV analysis of seismic sections to identify bypassed pay zones and structural traps.

Stage 6 SPT Projection
MCDA ranking of wells for SPT treatment. Evaluates 8+ parameters.

Stage 7 Economic Analysis
NPV, IRR, payback period, Monte Carlo (10,000 iterations). Quantum-enhanced QAE.

Stage 8 Geophysical Expertise
Well log interpretation following Schlumberger methodology. LAS 2.0 import.

Stage 9 EOR Optimization
Final aggregation of all 8 prior stages. AI generates treatment recommendations prioritizing SPT.

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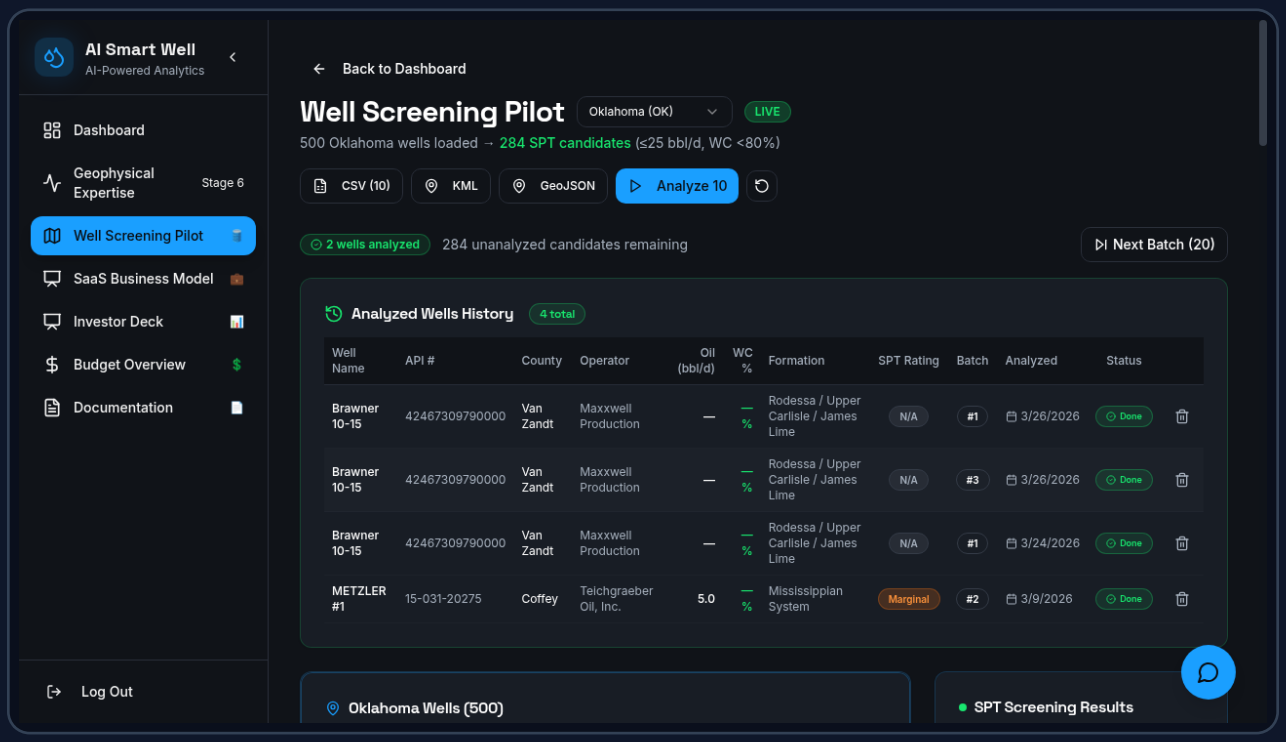

Figure 2 — Well Screening Pilot: Oklahoma 500-well analysis with SPT candidate ranking

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3. Key Functional Modules

Well Screening Pilot

Interactive pilot for multi-state well screening with well map, log visualization, and AI-powered stage processing.

Well Analysis Pipeline

End-to-end 9-stage execution engine with real-time progress and data transparency badges.

Reserves Map

Interactive Leaflet map showing remaining IOIP distribution with color-coded markers.

Production History

CSV import and visualization of monthly production data with decline curve overlay.

Data Import

Multi-source data ingestion: CSV, manual entry, API integration with well deduplication.

Geological Analysis

3D model, cross-section, well log display, anomaly detection, bypassed reserves identification.

AI Analyst (Chat)

Natural language AI consultant powered by Claude with MCDA and SPT context.

Automation Center

Real-time monitoring with configurable alert triggers and auto-linking.

Fluid Physics Simulation

Canvas-based interactive reservoir simulation with oil/water/gas particles.

Financial Calculator

ROI, NPV, and IRR calculations for intervention planning.

Real-Time Dashboard

Streaming metrics with WebSocket updates, production KPIs, and alert feed.

NVIDIA Cosmos

Three AI modules — Predict, Transfer, Reason — powered by NVIDIA's platform.

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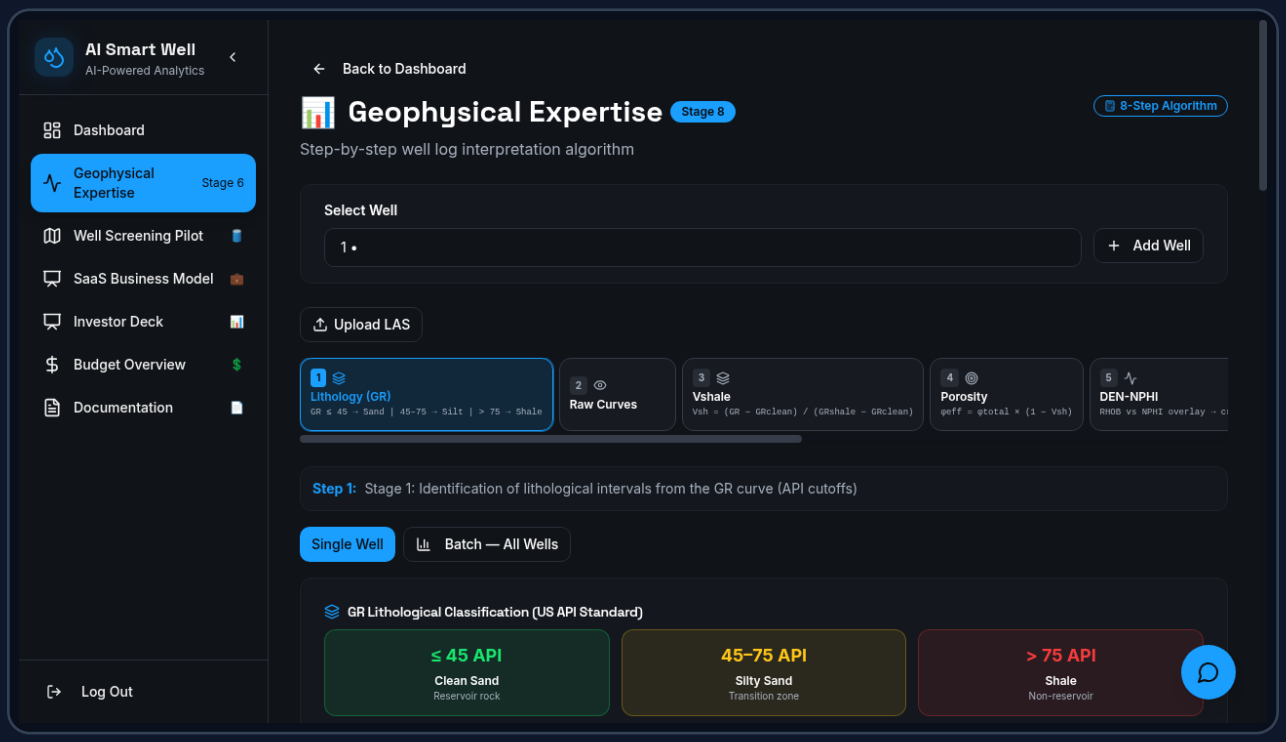

Figure 3 — Geophysical Expertise (Stage 8): Schlumberger petrophysical workflow

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4. Economic Analysis Module (Stage 7)

The Economic Analysis module provides comprehensive financial evaluation of SPT treatment candidates, combining deterministic NPV calculations with stochastic Monte Carlo simulation and quantum-enhanced risk assessment. This is a critical decision-support tool for operators evaluating well intervention economics.

Key Financial Metrics

NPV	IRR	PI	DPP
Net Present Value	Internal Rate of Return	Profitability Index	Discounted Payback

Monte Carlo Simulation Engine

- **10,000 iterations** — Full stochastic sampling across all uncertain parameters
- **8 variable distributions:** oil price, production rate, decline rate, OPEX, CAPEX, water cut, discount rate, treatment cost
- **Output:** NPV probability distribution with P10/P50/P90 confidence intervals
- **Visualization:** Interactive histogram with cumulative probability overlay

Tornado Sensitivity Analysis

Automated tornado chart identifies the parameters with the highest impact on project NPV. Each parameter is varied ±20% from its base case to measure sensitivity. Results are ranked by absolute impact, enabling operators to focus risk mitigation on the most critical variables.

Configurable Economic Parameters

Parameter	Default Range	Distribution
Oil Price (WTI)	$55–85/bbl	Log-normal
Initial Production Rate	5–50 bbl/d	Triangular
Annual Decline Rate	8–25%	Uniform
Operating Cost (OPEX)	$8–18/bbl	Normal
SPT Treatment Cost	$50K–200K	Triangular
Water Cut	20–85%	Beta
Discount Rate	8–15%	Fixed/Uniform
Tax Rate	15–35%	Fixed

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Quantum-Enhanced Analysis (QAE)

The Quantum Amplitude Estimation module provides quadratic speedup over classical Monte Carlo. Using 100 quantum iterations achieves the same precision as 10,000 classical iterations, with confidence intervals of ±0.8% vs classical ±5.2%.

Export & Reporting

- Full Monte Carlo report export (PDF) with P10/P50/P90 summary
- Tornado chart export with parameter sensitivity ranking
- Comprehensive economic report integrating all financial metrics
- CSV data export for external financial modeling

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5. Geological & Geophysical Analysis

The Geological Analysis module is the platform's core interpretation engine, combining seismic reinterpretation (Stage 5), well log analysis (Stage 8), and 3D geological modeling into a unified AI-powered workspace. It integrates NVIDIA NIM computer vision for automated feature detection.

Module Components

Seismic Amplitude Analysis

- Interactive seismic section display with 3 amplitude traces
- 4 interpreted horizons: Top Reservoir, Base Sandstone, Fault Zone, Basement
- Attribute analysis: Coherence (0.85), Dip Angle, Interval Velocity, AVO Gradient
- CSV upload for real operator seismic data (replaces synthetic demo)
- AI interpretation via Gemini — generates expert geological reports

Well Selector Integration

- Link any well from the database to seismic analysis context
- Auto-populates formation, depth, county, operator, production data
- AI analysis incorporates well-specific parameters for targeted interpretation

Auto-Classification Engine

- Seismic facies auto-classification: Channel Sand, Reef Buildup, Salt Dome, Shale Drape, Carbonate Platform
- Confidence scoring for each detected facies type
- Pattern-based anomaly detection with threshold alerting

Seismic Image CV Analysis (NVIDIA NIM)

- Three analysis modes: Horizon Detection, Fault Network Mapping, Amplitude Anomaly
- Upload seismic section images for AI computer vision processing
- Automated identification of structural traps and bypassed pay zones
- Results stored in seismic_analyses table with full audit trail

Bypassed Reserves Identification

- AI calculates estimated IOIP in bypassed zones
- Recovery factor estimation based on reservoir properties
- Economic screening of identified reserves against SPT treatment cost

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5. Geological Analysis (continued)

Well Cross-Section View

Interactive geological cross-section displaying layered formation structure with well paths. Six formation layers are visualized: Topsoil, Sandstone, Limestone, Shale (target reservoir), Dolomite, and Basement. Wells are shown with depth indicators and perforation intervals.

3D Geological Model

- Interactive 3D visualization of subsurface geology
- Formation boundaries rendered with depth-color mapping
- Well trajectory overlay with deviation surveys
- Rotatable/zoomable canvas for detailed structural analysis

Well Log Visualization

- Multi-track display: GR, Resistivity, Porosity, Sw, Density, Neutron Porosity
- Lithology track with color-coded intervals (Sand/Silt/Shale)
- Perforation intervals (PERF track) from well_perforations table
- LAS 2.0 file import with standard mnemonic recognition
- Schlumberger 9-step petrophysical workflow integration

Seismic Horizons Map

Interactive Leaflet-based map showing seismic survey coverage over the Permian Basin. Color-coded horizon contours display structural depth variations across the survey area. Well locations are plotted with pop-up details including API number, formation, and production data.

Analysis Progress Dashboard

847	45	78%	3
Wells Analyzed	Formations Mapped	Overall Progress	CV Analysis Modes

Database Tables Used

Table	Role in Geological Analysis
wells	Well coordinates, formation, depth, status for context
well_logs	GR, resistivity, porosity, Sw, density measurements

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Table	Role in Geological Analysis
well_perforations	Completion intervals and perforation details
seismic_images	Uploaded seismic section images for CV analysis
seismic_analyses	AI interpretation results with mode and confidence
core_images	Rock sample photos for CV-based lithology classification
core_analyses	Text and CV analysis results for core samples

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6. NVIDIA Cosmos Integration

NVIDIA Cosmos is deeply integrated into the AI Smart Well platform, providing three specialized AI modules that enhance the 9-stage analysis pipeline with physics-grounded simulation, synthetic data generation, and explainable AI reasoning. All three modules leverage NVIDIA's world foundation models for physically accurate predictions.

6.1 Cosmos Predict — Reservoir Simulation

Cosmos Predict simulates post-SPT reservoir behavior using real well log data. It generates an 18-month production forecast showing the expected impact of Slot Perforation Technology on well performance.

Key Capabilities

- Uses real well logs (e.g., Brawner 10-15) via useWellLogs hook — displays 'REAL DATA' badge
- 18-month production forecast with pre-treatment vs post-treatment comparison
- Movable SPT treatment zone — operator can adjust treatment depth interactively
- Physics-grounded simulation using NVIDIA world foundation models
- Preset well profiles for demo: high-production, marginal, depleted scenarios
- Integration with Stage 6 (SPT Projection) for candidate evaluation

Parameter	Source	Usage
Gamma Ray (GR)	Well logs / LAS import	Lithology identification for pay zone selection
Resistivity (RT)	Well logs / LAS import	Fluid saturation estimation
Neutron Porosity (NPHI)	Well logs / LAS import	Porosity and gas effect detection
Formation Properties	FORMATION_DB (200+)	Permeability, porosity baseline ranges
Production History	production_history table	Decline curve baseline for forecast
SPT Zone Depth	User-configurable	Treatment interval selection (interactive slider)

6.2 Cosmos Transfer — Synthetic Data Generation

Cosmos Transfer addresses the critical challenge of data-poor basins by generating synthetic well log data that closely matches real measurements. This enables analysis of wells in regions with limited historical data, dramatically expanding the platform's geographic coverage.

Key Capabilities

- Synthetic generation of GR, RT, and NPHI curves for data-poor wells
- Model accuracy improvement: **58% to 91%** with transfer learning
- Reference well: Brawner 10-15 (API 42-467-30979), East Texas Basin
- Style transfer from data-rich basins (Permian, Eagle Ford) to target regions
- Quality validation — synthetic vs real comparison metrics
- Seamless integration with Tier 2 (Formation-Based) data fallback strategy

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Metric	Before Transfer	After Transfer
Log Curve Correlation	58%	91%
Porosity RMSE	4.2 p.u.	1.1 p.u.
Lithology Classification	62%	89%
Pay Zone Identification	55%	87%

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6. NVIDIA Cosmos Integration (continued)

6.3 Cosmos Reason — Explainable AI Ranking

Cosmos Reason uses Chain-of-Thought (CoT) methodology to provide transparent, step-by-step well candidate ranking. Unlike black-box models, every ranking decision is fully traceable with documented reasoning at each step.

Key Capabilities

- Chain-of-Thought (CoT) reasoning with progressive step visualization
- 6-metric radar chart: Porosity, Permeability, Depth, Production, Water Cut, GOR
- All metrics normalized (0-100); Water Cut and GOR **inverted** (higher = better)
- Radar chart unfolds progressively as each reasoning step completes
- Full reasoning transcript available for audit and regulatory compliance
- Integration with Stage 6 (SPT Projection) MCDA ranking for cross-validation

Step	Action	Output
1. Data Ingestion	Load well logs, production, completion data	Structured feature vector
2. Feature Extraction	Calculate porosity, permeability, Sw from logs	6 normalized metrics (0-100)
3. Constraint Check	Verify depth, water cut, GOR thresholds	Pass/fail for each criterion
4. Comparative Analysis	Rank against peer wells in same formation	Percentile position
5. SPT Suitability	Evaluate slot perforation potential	SPT score (0-100)
6. Final Recommendation	Aggregate scores, generate reasoning text	Ranked candidate list

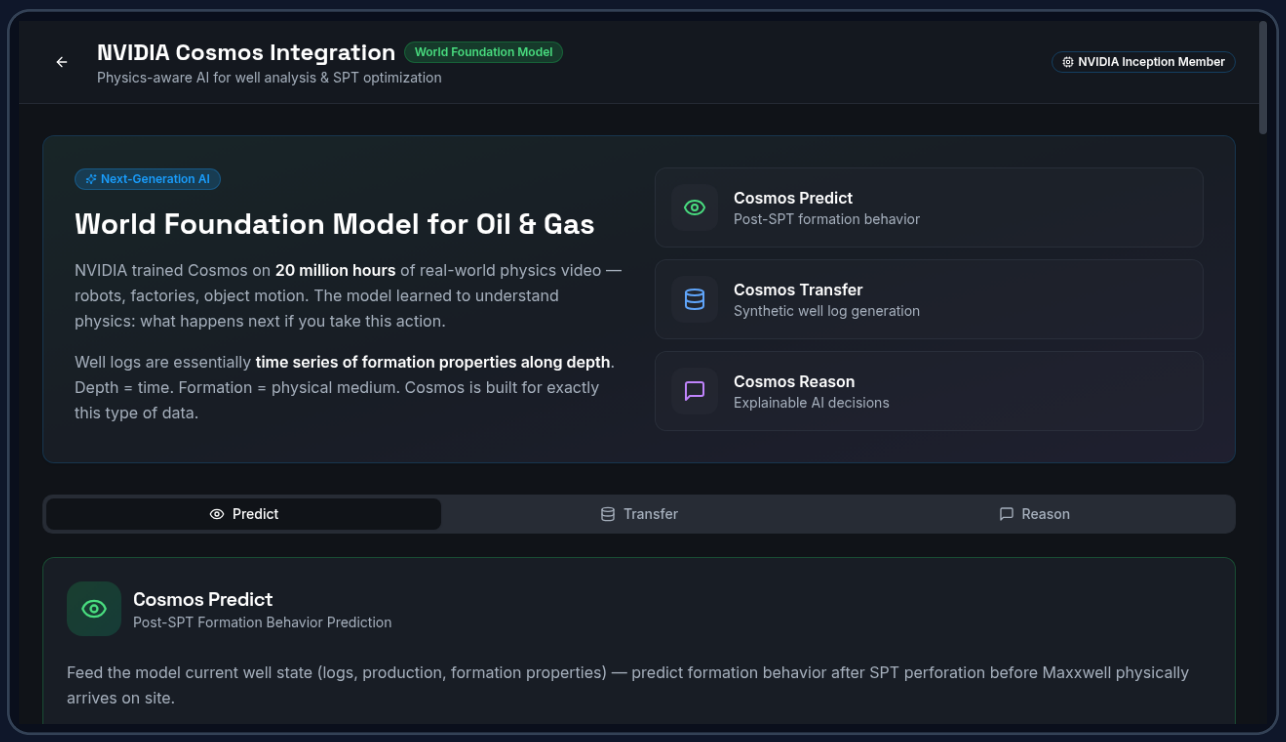

Figure 4 — NVIDIA Cosmos: Predict, Transfer, and Reason modules in action

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NVIDIA Partnership

- **NVIDIA Inception Program** — Accepted startup member with access to DGX Cloud credits
- **NVIDIA NIM** — Nemotron Nano 12B v2 VL for computer vision (core + seismic)
- **NVIDIA Cosmos** — World foundation models for physics-grounded simulation
- **AWS Activate** — $100K cloud credits for GPU-accelerated inference workloads

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7. AI & Machine Learning Capabilities

Computer Vision (NVIDIA NIM)

Model: Nemotron Nano 12B v2 VL via NVIDIA API Catalog

- Core sample analysis: porosity estimation, fracture detection, mineralogy classification
- Seismic section interpretation: horizon picking, fault detection, amplitude anomalies
- Three specialized CV modes with configurable confidence thresholds

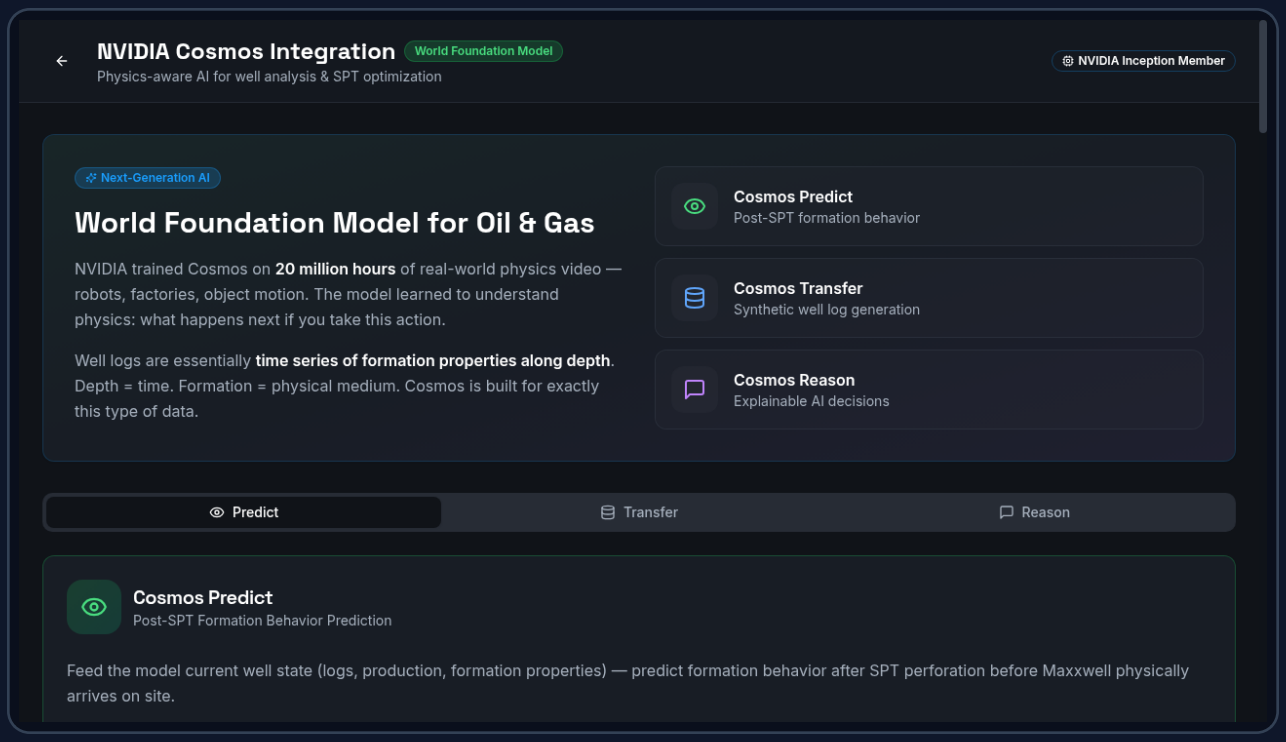

Figure 4 — NVIDIA Cosmos Integration: Predict, Transfer, and Reason modules

Natural Language AI

- Gemini 2.5 Flash — geological expert reports, well evaluations, stage analyses
- Claude — SPT AI Consultant chat with full well context
- Reliability: 3x retry with exponential backoff on all AI requests

Data Strategy

Three-tier data fallback hierarchy:

- **Tier 1 — Real Data:** Imported logs, production history, core images
- **Tier 2 — Formation-Based:** Values from FORMATION_DB (200+ formations)
- **Tier 3 — Synthetic:** Deterministic seeded random walk (500 points, stableHash)

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8. Quantum Monte Carlo Engine

The platform includes a Quantum-Enhanced Monte Carlo simulation engine that leverages Quantum Amplitude Estimation (QAE) for accelerated economic risk analysis.

How It Works

- **Step 1 — State Preparation:** Encode probability distributions into quantum state superposition
- **Step 2 — Amplitude Amplification:** Apply Grover-like rotations to amplify high-probability states
- **Step 3 — Measurement:** Extract NPV distribution with quadratic speedup O(1/N) vs classical O(1/sqrt(N))

Performance Comparison

Parameter	Classical MC	Quantum MC
Iterations for 1% precision	10,000	100
Convergence rate	O(1/sqrt(N))	O(1/N)
Confidence interval	+-5.2%	+-0.8%
Processing approach	Sequential sampling	Superposition encoding

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9. Data Management & Security

Database Schema (14 Tables)

Table	Purpose
wells	Master well records (API, coordinates, formation, production, status)
well_logs	Petrophysical measurements (GR, resistivity, porosity, Sw, density)
well_perforations	Completion data (depth intervals, shots/ft, phasing, status)
well_analyses	9-stage pipeline results with batch tracking
well_alerts	Real-time monitoring alerts (severity, thresholds, read status)
production_history	Monthly oil/gas/water production by well
core_images	Core sample photos linked to wells and formations
core_analyses	AI interpretation results for core samples
seismic_images	Seismic section images with metadata
seismic_analyses	CV and AI interpretation of seismic data
formation_codes	State-specific formation code reference (including KS KID)
companies	Tenant organizations
user_companies	User-to-company membership mapping
user_roles	RBAC role assignments (admin, investor)

Security Architecture

- **Multi-Tenant Isolation:** Row-Level Security (RLS) on all data tables via company_id
- **RBAC:** Role-based access (admin/investor) with security definer functions
- **Authentication:** Email/password + Google OAuth, registration disabled (invite-only)
- **API Security:** JWT validation on all backend functions

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10. Visualization & Simulation

Interactive Visualizations

- Well log display with GR, resistivity, porosity, Sw, and lithology tracks
- Geological cross-section with interactive well selection
- Decline curve charts with Arps model overlay (Recharts)
- Monte Carlo histograms with P10/P50/P90 markers
- Tornado sensitivity charts for economic parameter impact
- Reserves map with Leaflet — color-coded by IOIP volume

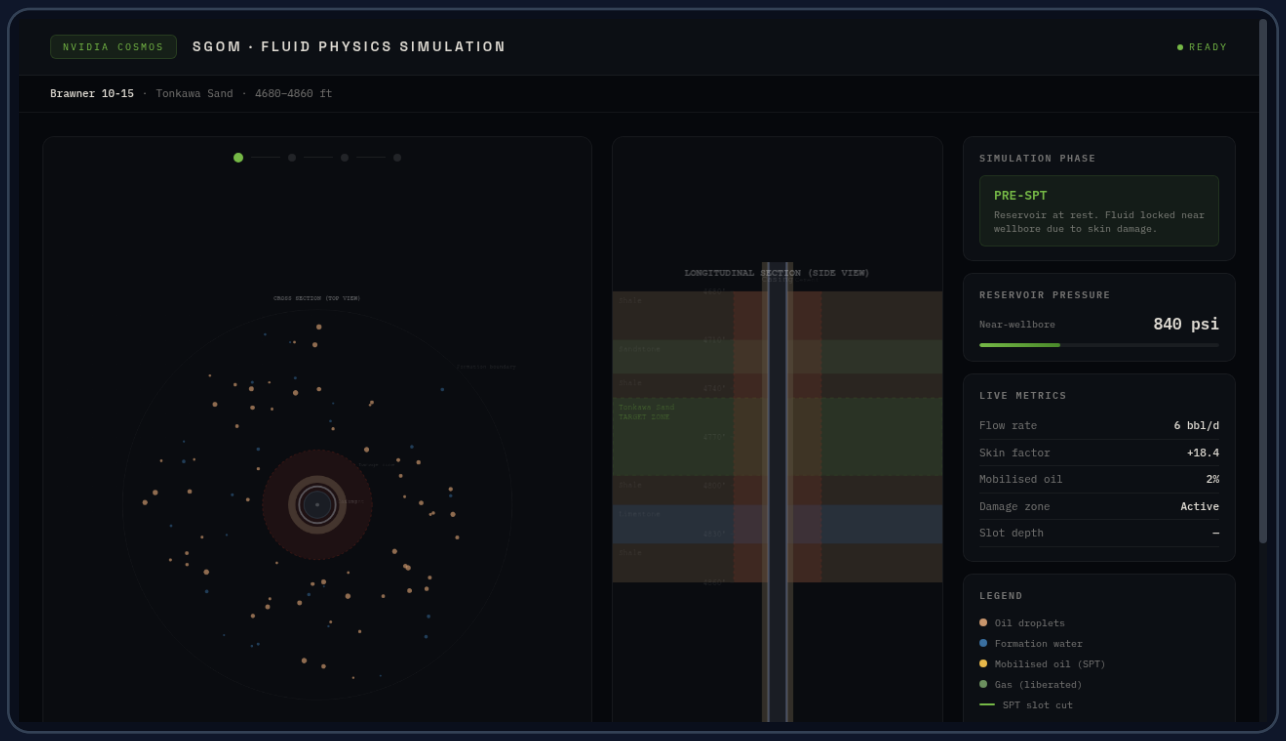

Figure 5 — Fluid Physics Simulation: Real-time reservoir particle dynamics with SPT treatment phases

Physics Simulations

- Canvas-based fluid particle simulation (oil, water, gas phases)
- Injection/production response visualization
- Formation boundary and permeability zone effects
- Completion visualization — wellbore cross-section animation

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11. Technology Stack

Layer	Technology	Purpose
Frontend	React 18, TypeScript 5, Vite 5	UI framework and build system
Styling	Tailwind CSS v3, shadcn/ui	Design system with dark theme
Charts	Recharts	Production/economic data visualization
Maps	Leaflet + React-Leaflet	Well location and reserves mapping
State	TanStack Query	Server state management and caching
Backend	Supabase (PostgreSQL 15)	Database, auth, real-time, storage
Functions	Deno Edge Functions (x12)	AI proxy, data fetching, analysis
AI Vision	NVIDIA NIM (Nemotron Nano 12B)	Core/seismic image analysis
AI Text	Gemini 2.5 Flash, Claude	Expert reports and chat assistant
Video	Remotion + Puppeteer	Programmatic animation rendering
Security	RLS, JWT, RBAC	Multi-tenant data isolation

Backend Functions (12 Edge Functions)

- **analyze-core / analyze-core-cv** — AI core sample interpretation (text + CV)
- **analyze-seismic / analyze-seismic-cv** — Seismic section AI analysis
- **analyze-well-stage** — Pipeline stage processing
- **rank-wells** — MCDA well ranking
- **spt-chat** — AI consultant chat proxy (Claude)
- **fetch-wells / fetch-texas-wells** — State registry API
- **fetch-nearby-wells** — Bounding box nearby well search
- **lookup-well-by-api** — Single well lookup
- **get-oil-price** — Real-time oil price feed

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12. Business Model & Pricing

SaaS Subscription Tiers

Tier	Price	Wells	Features
Explorer	$2,000/mo + $120/well	Up to 50	Core pipeline, basic reports
Professional	$5,000/mo + $95/well	Up to 500	Full pipeline + Quantum MC, API access, priority support
Enterprise	$12,000/mo + $75/well	Unlimited	Custom models, dedicated infrastructure, SLA

Revenue Projections

- Year 1: 15 clients x avg $4,500/mo = **$810K ARR**
- Year 2: 45 clients x avg $5,200/mo = **$2.8M ARR**
- Year 3: 120 clients x avg $6,000/mo = **$8.6M ARR**

MVP development budget: **$133,000**

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13. Competitive Advantages

✓ **Patented SPT Technology**

US Patent 8,863,823 — 5-10x inflow increase. No competing platform integrates patented completion technology.

✓ **AI-First Architecture**

Every stage is AI-augmented. NVIDIA NIM for CV, Gemini for reports, Claude for consultation.

✓ **Quantum Computing Edge**

QAE delivers quadratic speedup. 100 quantum iterations = 10,000 classical iterations.

✓ **9-Stage Standardized Pipeline**

Repeatable, auditable methodology. Competitors offer point solutions; we deliver end-to-end.

✓ **Multi-State Registry Integration**

Direct API access to well registries across 7 states.

✓ **Enterprise Security by Design**

RLS on every table. Multi-tenant isolation. RBAC. Invite-only access.

✓ **Formation Intelligence Database**

200+ pre-calibrated formation property ranges for instant analysis.

✓ **Cost-Efficient Data Strategy**

Users bring their own data API keys. Platform provides the intelligence layer.

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